

Whois & DNS Fingerprinting

1. Whois

Whois is a protocol used to query databases that store registered users or assignees of domain names and IP addresses. It provides information like:

- Domain registrant (person/organization)
- Registrant's contact information
- Nameservers associated with the domain
- Domain registration dates
- Expiry dates

Whois tools can be used to obtain information about a domain and its ownership. Some popular Whois tools include:

- whois command in Linux
- Online services like whois.domaintools.com or whois.icann.org

2. DNS Fingerprinting

DNS fingerprinting refers to techniques used to gather information about DNS servers and the way they handle DNS queries. This can reveal useful information about the infrastructure behind a domain or network.

Common techniques:

- **DNS Query Fingerprinting:** Identifying the DNS server type or software by analyzing the responses to specific queries.
- **DNS Interrogation:** Sending specific types of DNS queries to a target and analyzing how the server responds to infer server configurations.
- **Tool-based Fingerprinting:** Tools like `dnsmmap`, `dnscan` DNS scripts can help detect DNS server software.

Using whois CLI in Kali Linux:

```
(kali@kali)-[~/burp]
$ whois serverfault.com
Domain Name: SERVERFAULT.COM
Registry Domain ID: 1547166245_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.corporatedomains.com
Registrar URL: http://cscdbs.com
Updated Date: 2025-03-11T05:21:33Z
Creation Date: 2009-03-15T03:13:53Z
Registry Expiry Date: 2026-03-15T03:13:53Z
Registrar: CSC Corporate Domains, Inc.
Registrar IANA ID: 299
Registrar Abuse Contact Email: domainabuse@cscglobal.com
Registrar Abuse Contact Phone: 8887802723
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Name Server: DAMIAN.NS.CLOUDFLARE.COM
Name Server: SUREENA.NS.CLOUDFLARE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2025-03-25T08:23:44Z <<<
```

This list out many details such as registrar details, creation data, updated date, last update of whois database etc.

Also this tool provide many details which is useful for forensic experts to gain information

```
Domain Name: serverfault.com
Registry Domain ID: 1547166245_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.corporatedomains.com
Registrar URL: www.cscprotectsbrands.com
Updated Date: 2025-03-11T01:21:33Z
Creation Date: 2009-03-14T23:13:53Z
Registrar Registration Expiration Date: 2026-03-15T03:13:53Z
Registrar: CSC CORPORATE DOMAINS, INC.
Sponsoring Registrar IANA ID: 299
Registrar Abuse Contact Email: domainabuse@cscglobal.com
Registrar Abuse Contact Phone: +1.8887802723
Domain Status: clientTransferProhibited http://www.icann.org/epp#clientTransferProhibited
Registry Registrant ID:
Registrant Name: Sysadmin Team
Registrant Organization: Stack Exchange, Inc.
Registrant Street: 14 Wall Street, 20th Floor
Registrant City: New York
Registrant State/Province: NY
Registrant Postal Code: 10005-2123
Registrant Country: US
Registrant Phone: +1.2122328280
Registrant Phone Ext:
Registrant Fax: +1.2122328280
Registrant Fax Ext:
Registrant Email: sysadmin-team@stackoverflow.com
Registry Admin ID:
Admin Name: Sysadmin Team
Admin Organization: Stack Exchange, Inc.
Admin Street: 14 Wall Street, 20th Floor
Admin City: New York
Admin State/Province: NY
Admin Postal Code: 10005-2123
Admin Country: US
Admin Phone: +1.2122328280
Admin Phone Ext:
Admin Fax: +1.2122328280
Admin Fax Ext:
Admin Email: sysadmin-team@stackoverflow.com
Registry Tech ID:
Tech Name: Sysadmin Team
Tech Organization: Stack Exchange, Inc.
Tech Street: 14 Wall Street, 20th Floor
Tech City: New York
Tech State/Province: NY
Tech Postal Code: 10005-2123
Tech Country: US
Tech Phone: +1.2122328280
Tech Phone Ext:
Tech Fax: +1.2122328280
```

What is Snort?

Snort is the foremost Open Source Intrusion Prevention System (IPS) that detects and prevents malicious network activity and uses those rules

Using Whois.com

The screenshot shows the Whois.com website interface. At the top, there's a navigation bar with links: Domains, Hosting, Servers, Email, Security, Whois, Deals. A search bar contains 'serverfault.com'. The main content area is titled 'serverfault.com' and 'Updated 1 second ago'. It features three main sections: Domain Information, Registrar Information, and Registrant Contact. To the right, there's a list of 'Interested in similar domains?' with options like server-fault.com, serversfault.com, dnsfault.com, etc. Below that, there are two promotional banners: one for '.space' domains at \$1.18 and another for '.info' domains at \$4.98.

Domain Information

Domain:	serverfault.com
Registered On:	2009-03-15
Expires On:	2026-03-15
Updated On:	2025-03-11
Status:	client transfer prohibited
Name Servers:	damian.ns.cloudflare.com sureena.ns.cloudflare.com

Registrar Information

Registrar:	CSC Corporate Domains, Inc.
IANA ID:	299
Abuse Email:	domainabuse@cscglobal.com
Abuse Phone:	8887802723

Registrant Contact

Name:	Sysadmin Team
Organization:	Stack Exchange, Inc.
Street:	14 Wall Street, 20th Floor
City:	New York
State:	NY

Interested in similar domains?

- server-fault.com [Buy Now](#)
- serversfault.com [Buy Now](#)
- dnsfault.com [Buy Now](#)
- serverfaults.com [Buy Now](#)
- serverfault.net [Buy Now](#)
- serversfault.net [Buy Now](#)

.space Sale
\$29.88 **\$1.18**
[BUY NOW](#)
*while stocks last

On Sale! .info
.INFO @ \$4.98 ~~\$29.88~~
[BUY NOW](#)

Using who.is

The screenshot shows the who.is website interface. At the top, there's a navigation bar with links: WHOIS Search, Domain Name, Website, and IP Tools. A search bar contains 'www.geeksforgeeks.org'. Below the search bar, it says 'Your IP address is 103.127.187.246'. The main content area is titled 'who.is' and 'Website information and creation services'.

who.is
Website information and creation services

WHOIS Search, [Domain Name](#), Website, and IP Tools

[www.geeksforgeeks.org](#)

Your IP address is [103.127.187.246](#)

The output it gives is:

The screenshot shows the geeksforgeeks.org website interface. At the top, there's a navigation bar with links: Whois, RDAP, DNS Records, Uptime, Diagnostics. The main content area is titled 'geeksforgeeks.org' and 'whois information'. It features a 'Chat with an expert Online Now' button. Below that, there's a table showing 'Registrar Info' and 'Site Status'. To the right, there's a list of 'Suggested Domains for geeksforgeeks.org'.

geeksforgeeks.org
whois information

[Whois](#) [RDAP](#) [DNS Records](#) [Uptime](#) [Diagnostics](#)

Registrar Info

Name	PDR Ltd. d/b/a PublicDomainRegistry.com
Whois Server	http://whois.publicdomainregistry.com
Referral URL	http://www.publicdomainregistry.com
Status	clientTransferProhibited https://icann.org/epp#clientTransferProhibited

Site Status

Status	Active
Server type	nginx

Suggested Domains for geeksforgeeks.org

- ☐ geeks-for-geeks.live \$3.99
- ☐ geeksformord.live \$3.99
- ☐ nerdforgeeks.live \$3.99
- ☐ geeksforreak.live \$3.99
- ☐ breakforgeeks.live \$3.99

[Purchase Selected Domains](#)

Important Dates	
Expires On	2030-03-19
Registered On	2009-03-19
Updated On	2022-04-21
Name Servers	
ns-1520.awsdns-62.org	205.251.197.240
ns-869.awsdns-44.net	205.251.195.101
ns-245.awsdns-30.com	205.251.192.245
ns-1569.awsdns-04.co.uk	205.251.198.33
Similar Domains	
geeks-2-go.com geeks-2-you.com geeks-24.com geeks-24.info geeks-247.com geeks-2go.com geeks-360.com geeks-360.net geeks-3d.com geeks-3d.ru geeks-3d1.ru geeks-4-good.net geeks-4-good.org geeks-4-hire.com geeks-4-jesus.org geeks-4-less.com geeks-4-rent.com geeks-4-u.com geeks-411.com geeks-4u.com.au	
Registrar Data Make Private Now We will display stored WHOIS data for up to 30 days.	
Registrant Contact Information: Name: REDACTED FOR PRIVACY Organization: Privacy Protect, LLC (PrivacyProtect.org) Address Line 1: REDACTED FOR PRIVACY Address Line 2: City: REDACTED FOR PRIVACY State/Province: MA Postal Code: REDACTED FOR PRIVACY Country: US Phone: REDACTED FOR PRIVACY Fax: REDACTED FOR PRIVACY Email: Please query the RDDS service of the Registrar of Record identified in this output for information on how to contact the Registrant, Admin. or Tech contact of the queried domain name.	

Using dnsmap

```
(kali㉿kali)~[~/burp]
$ dnsmap serverfault.com
dnsmap 0.36 - DNS Network Mapper

[+] searching (sub)domains for serverfault.com using built-in wordlist
[+] using maximum random delay of 10 millisecond(s) between requests

blog.serverfault.com
IP address #1: 198.252.206.19

chat.serverfault.com
IP address #1: 172.64.148.218
IP address #2: 104.18.39.38

meta.serverfault.com
IP address #1: 104.18.39.38
IP address #2: 172.64.148.218

www.serverfault.com
IP address #1: 104.18.39.38
IP address #2: 172.64.148.218

[+] 4 (sub)domains and 7 IP address(es) found
[+] completion time: 129 second(s)
```

This tool is useful for dns fingerprinting, which shows the ip address of the domain.

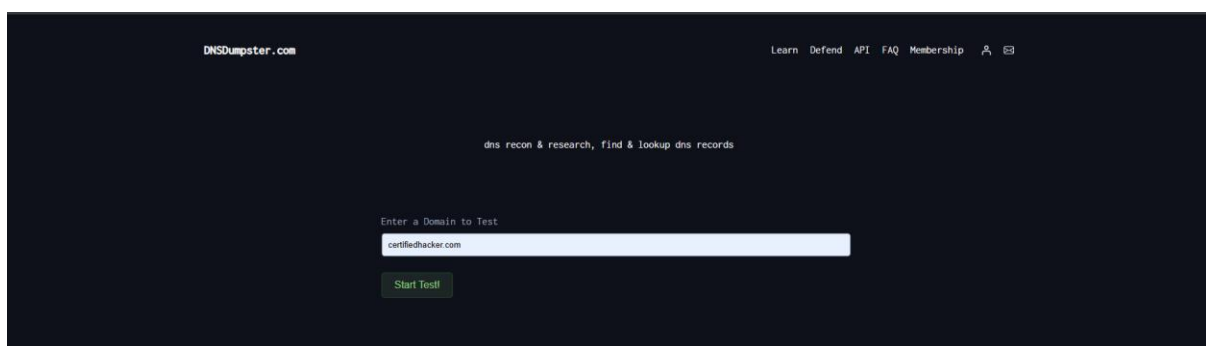
Nslookup

```
(kali㉿kali)-[~/burp]
$ nslookup serverfault.com
Server:      192.168.1.1
Address:     192.168.1.1#53

Non-authoritative answer:
Name:   serverfault.com
Address: 172.64.148.218
Name:   serverfault.com
Address: 104.18.39.38
```

This tool is helpful to obtain domain name or IP address information. It helps in resolving domain names to IP addresses (or vice versa) and troubleshooting DNS-related issues.

DnsDumpster



The output is :

